

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 1.4 TRANSPORT IN PLANTS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the driving question: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves, and what happens to that water in the end? Think back to the two puzzling observations in Lesson 1: the red dye that travelled through a maize seedling from the Kisumu school garden, and the water droplets that formed inside the sealed polythene bag around a sweet potato cutting in afternoon sun. Answer every section in the blank space under each prompt. Use full sentences and scientific words from our lessons. Draw labelled diagrams where a prompt asks for one. Link each answer to the evidence we collected in class. Use your own thinking and what you observed.

Part 1: Why the dye appeared only in certain places

In Lesson 1, a maize seedling from the Kisumu school garden stood with its roots in red-dyed water for 30 minutes. When the stem was cut across, vivid red rings appeared only in specific regions, while the outer green tissue and the central pith stayed white.

(a) Which tissue turned red? Explain how you know it is this tissue and not phloem.

(b) Describe two features of this tissue that help water move upward through it.

(c) Use the capillary tube demonstration to explain how adhesion and cohesion help water rise in narrow tubes.

(a) The red rings were the xylem. Xylem carries water and mineral salts upward from the roots. The dye was dissolved in the water the roots absorbed, so it travelled in the xylem stream and stained only those channels. Phloem carries dissolved sugars made in the leaves, not water taken up from the soil, so it stayed colourless, as did the pith and the outer tissue.

(b) Xylem vessels are narrow, hollow, dead cells joined end to end, so there is a continuous, open tube with no cell contents blocking the flow. Being very narrow helps water rise by capillary action.

(c) In the capillary demonstration, water rose higher in the narrower tubes. Adhesion is water sticking to the walls of the tube, and cohesion is water molecules sticking to each other. Adhesion pulls water up along the wall and cohesion pulls the rest of the water along with it as an unbroken column. Xylem vessels are narrow like the thin glass tubes, so the same forces help water rise in the maize stem.

Part 2: Getting water and minerals into the root

A farmer in the Rift Valley

(a) The cell sap in the root hair cell vacuole is more concentrated than the soil water, so it has a lower water potential. The soil

applies DAP fertiliser to her maize. (a) Explain how water enters a root hair cell, using what you learned about the cell sap and the soil water. (b) Explain how phosphate ions from the DAP enter the root hair cell. Say why this process is different from the way water enters. (c) Explain how the structure of a root hair cell suits its job. (d) What can happen to maize or sukuma wiki if too much fertiliser is applied? Explain why. (e) A maize plant in Mombasa has wilted in the afternoon heat but recovers after it is watered in the evening. Explain how the plant's cells recover.	water is dilute, with a higher water potential. Water moves by osmosis from the higher water potential (soil) to the lower water potential (cell sap) across the semi-permeable cell membrane. (b) Phosphate ions are absorbed by active transport. Protein carrier molecules in the cell membrane use energy from ATP to pump the ions into the cell against their concentration gradient. This is different from osmosis because osmosis needs no energy and moves water down a gradient, while active transport uses energy and moves ions against the gradient. (c) A root hair cell is a long, thin extension of the root cell. It gives a large surface area for absorbing water and mineral salts from the soil. (d) Too much fertiliser makes the soil water very concentrated, so its water potential becomes lower than that of the root cells. The gradient reverses and water leaves the root cells (plasmolysis), so the plant wilts. (e) When the soil is watered, the soil water is dilute again and has a higher water potential than the cell sap. Water enters the root cells by osmosis and the cells fill with water, restoring their turgor, so the plant recovers from wilting.
---	--

Part 3: Reaching the topmost leaves and carrying sugar	
A maize plant can be about two metres tall. (a) Root pressure pushes water into the xylem. Is this push enough to raise water to the topmost leaves of the plant? Explain, and describe what else is involved. (b) Name the tissue that carries sugars, state the process, and say where sugars travel from and to. (c) In the ring-barking (girdling) demonstration, tissue swelled just above the cut but not below. Explain what this shows and why a girdled stem eventually dies even though water still reaches its leaves.	(a) Root pressure is a push from the bottom, produced by water and ions accumulating in the root and xylem, and it is not strong enough to lift water to the top of a two-metre plant. Transpiration pull does most of the work. Water evaporates from the leaves, and because of cohesion the water molecules in the xylem are pulled up as a continuous column, with adhesion helping hold the water to the vessel walls (cohesion-tension). (b) Phloem carries sugars by translocation. Sugars move from sources, such as mature photosynthesising leaves, to sinks, such as the roots, the developing maize cob and young growing leaves. (c) Removing the ring of tissue cut through the phloem, so sugars coming down from the leaves piled up above the cut and caused swelling. Sugars could not pass below the cut. Xylem was left intact, so water still reached the leaves. However, the roots below the cut got no sugar and eventually died, and the whole plant then dies.

Part 4: What happens to the water in the end

Think about the sweet potato cutting sealed in the polythene bag in afternoon sun.

(a) Where did the water droplets inside the bag come from? Trace the journey of that water from the soil to the bag.

(b) In our group experiments (Groups A to D) we changed stomata, wind, humidity and light. Using the results your class recorded and graphed, describe how each factor affected the amount of water lost, and explain the pattern using the concentration gradient of water vapour.

(c) Give three reasons why transpiration is useful to the maize plant rather than just a loss of water.

(a) The water came from the plant. Root hair cells absorbed water from the soil by osmosis, and it travelled up through the xylem to the leaves. There it left through the stomata as water vapour by diffusion. The bag trapped the vapour, which then condensed on the cool inner surface of the bag as droplets. This process is transpiration.

(b) In our class results, the group with high wind lost the most water because wind sweeps away water vapour outside the stomata, keeping the concentration gradient steep. The group in bright light lost more than the group in shade because stomata open wider. The group in the sealed polythene bag lost very little because humidity rose, so the vapour concentration outside became high and the gradient flattened. The group with petroleum jelly blocking the stomata lost the least because vapour could not diffuse out.

(c) Transpiration (1) cools the leaf surface by evaporation on hot afternoons, (2) creates the transpiration pull that draws water up the xylem against gravity, and (3) maintains the transpiration stream that carries mineral salts, such as nitrogen and phosphorus from fertiliser, up to the topmost leaves.

Part 5: Putting it all together

Write a full explanation that answers the driving question: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves, and what happens to that water in the end?

Use a labelled diagram of a maize plant (roots, stem, leaf) with arrows for the movement of water, minerals and sugars. Then use your diagram in your written explanation. Finally, explain why transport is important to a plant. Add one piece of advice for a farmer in

Diagram: a maize plant with root hair cells at the root, a stem showing xylem and phloem, and a leaf with stomata. Arrows show water and minerals going up in the xylem, sugars moving in the phloem from the leaf (source) to the roots and cob (sinks), and water vapour leaving through the stomata.

Explanation: Water enters root hair cells by osmosis because the cell sap has a lower water potential than the dilute soil water. Mineral ions such as phosphate from DAP are taken in by active transport using ATP. Water and ions pass through the root to the xylem, where root pressure gives a push. The main force is transpiration pull: water evaporates from the leaves and, because of cohesion and adhesion in the narrow xylem vessels, a continuous column of water is pulled up to the topmost leaves, carrying dissolved minerals. This is why only the xylem rings turned red in our maize stem. In the leaf, the water leaves through the stomata as water vapour by diffusion, which is what formed the droplets in the polythene bag. Phloem separately moves sugars by translocation from source to sink.

Importance: Transport is important because every living cell needs water, minerals and sugars. Water and minerals are needed for photosynthesis and growth. Sugars from the leaves supply energy and materials to the roots, cob and growing parts. Transpiration also cools leaves and, from forests such as the Mau Forest, adds water vapour to the air for rainfall.

Advice: Water the crop regularly, especially in hot weather, and use manure and fertiliser such as DAP, CAN or Urea in the correct amounts, because too much fertiliser draws water out of the roots and makes the plant wilt.

Kisumu based on what you learned.	
-----------------------------------	--

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of the transport explanation	Correctly explains osmosis, active transport, root pressure, transpiration pull, xylem and phloem roles with all key terms used accurately.	Explains most processes correctly with minor errors or one missing mechanism.	Explanation has major errors or confuses xylem and phloem, or osmosis and active transport.
Use of evidence from the lessons	Consistently links claims to the red-dye maize stem, the polythene bag, the girdling demonstration and the transpiration group results.	Refers to some of the evidence, but links are sometimes unclear.	Little or no reference to class evidence.
Explaining transpiration and gradients	Explains the effect of wind, humidity, light and blocked stomata using the water vapour concentration gradient, and gives three uses of transpiration.	Describes the effect of the factors, with a partial explanation of the gradient or fewer uses.	Only lists results without a mechanism.
Linking to the driving question and Kenyan context	Gives a coherent whole-journey explanation, a clear labelled diagram, and sensible advice relating to Kenyan farming.	Gives a mostly complete journey with a diagram and some context.	Explanation is fragmented, with a missing or unclear diagram and no application.
Communication	Writes clearly in logical order with correct scientific vocabulary.	Writing is mostly clear with some vocabulary slips.	Writing is hard to follow and vocabulary is mostly everyday.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.